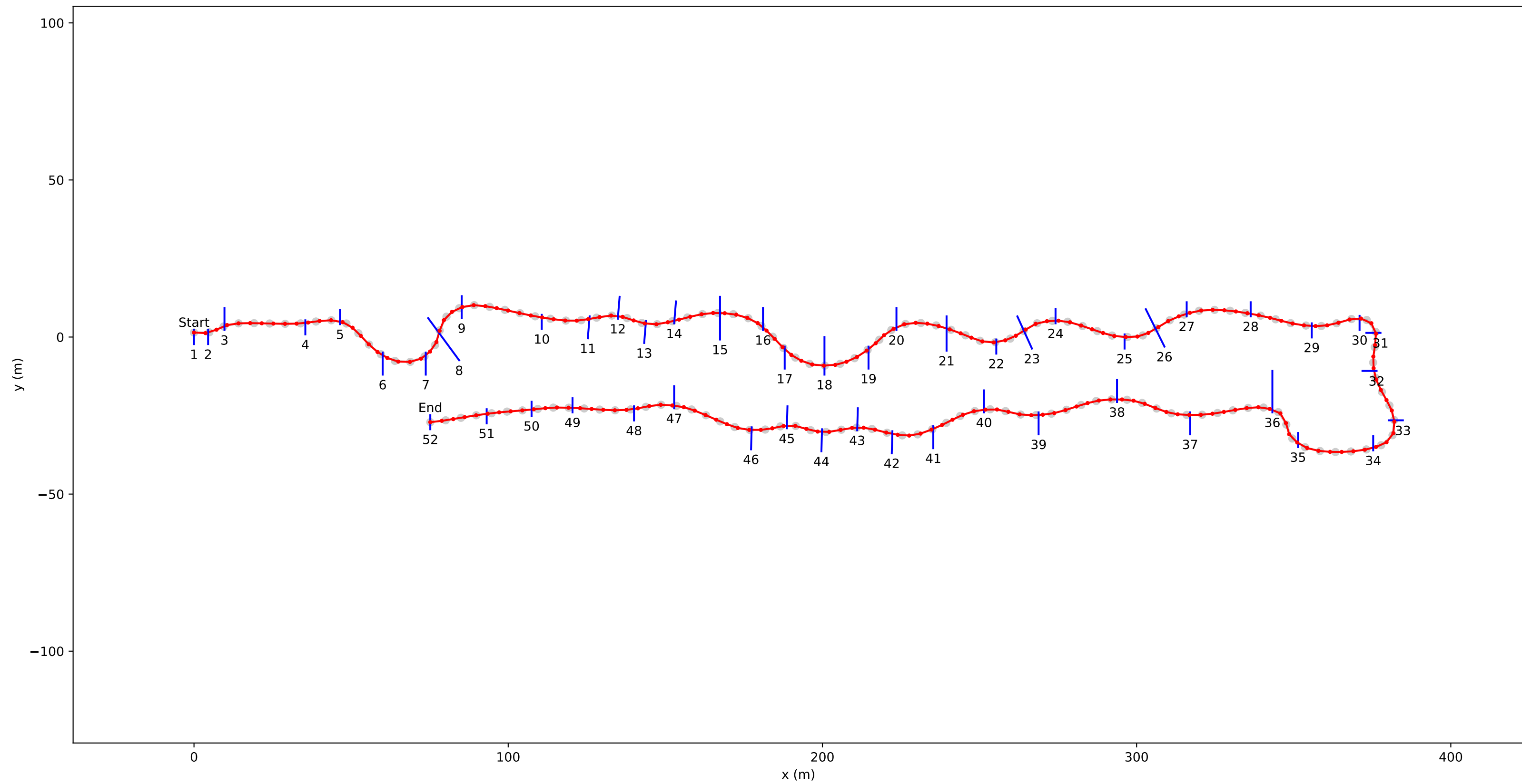


Map plot: optimized path through gates



Track Generation Report

=====
Generated: 2026-05-31 21:07:32
Gate file: ref_gates_autoX.csv
Gate count: 52
Saved DOFs file: opt_dofs_2026-05-31_21-07-32.npy

Optimization parameters

method: L-BFGS-B
maxiter: 50
ftol: 1e-07

Point counts

objective function points: 208
path generation points: 156

Results

total optimization time: 1574.953 s
total path length: 763.9283 m

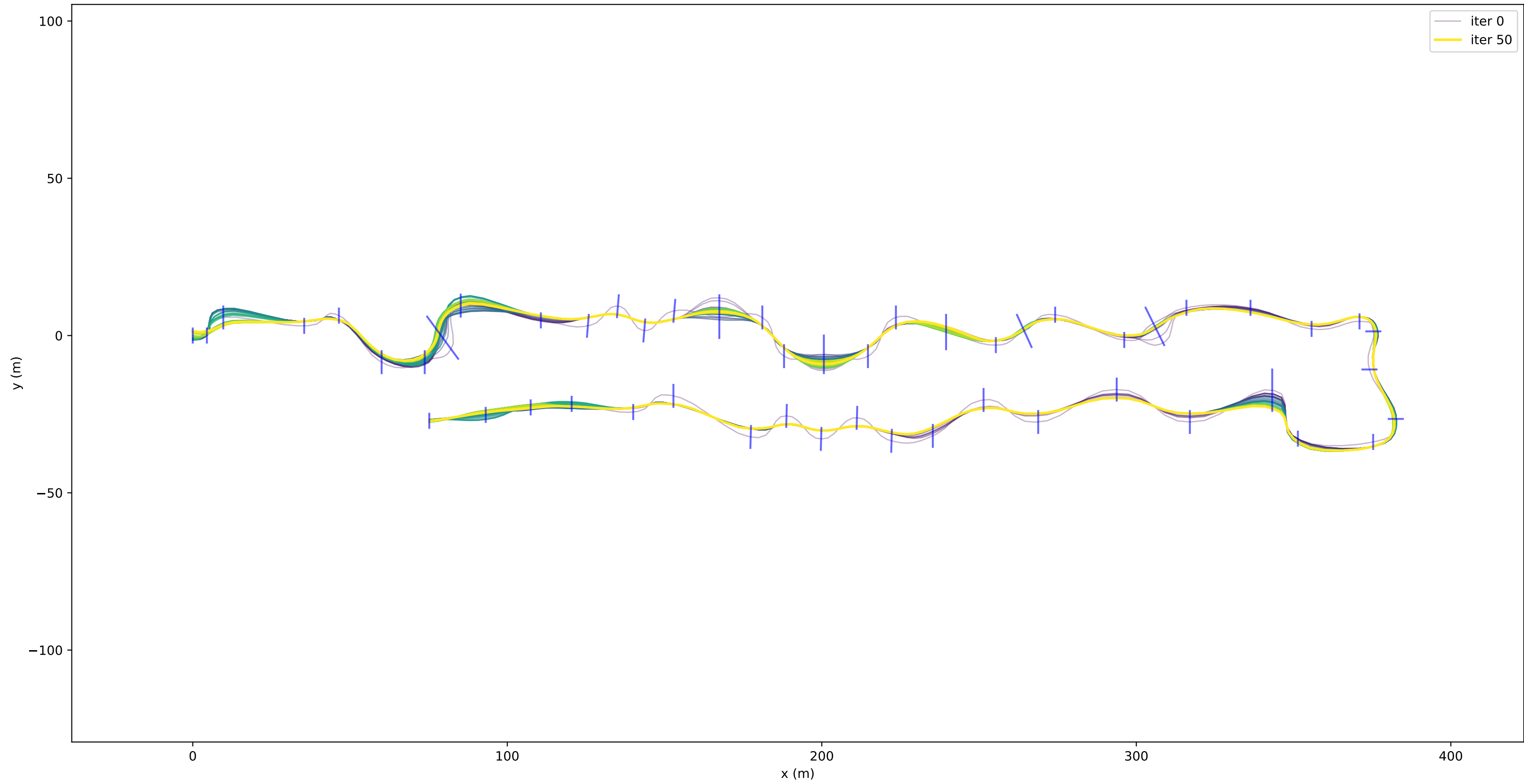
total objective function value: 2.377053

Per-step time estimate (one measured objective evaluation)

make spline from dofs: 6.24 ms
path length function: 0.49 ms
equidistant times generated: 0.30 ms
curvature objective function: 101.46 ms

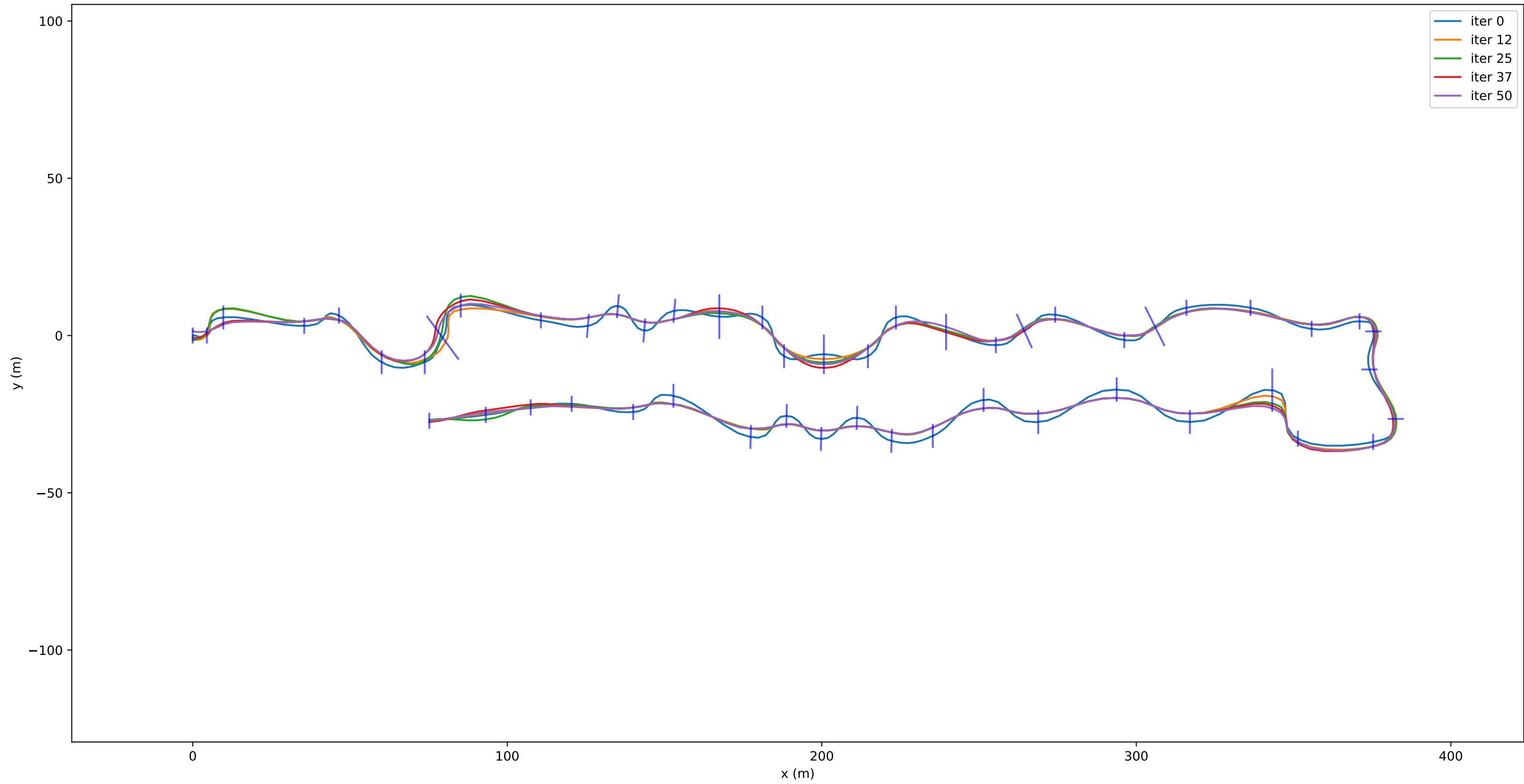
Optimization progression

Optimization progression (51 iterations)



Equally spaced iterations

5 equally spaced iterations



Instant radius vs distance

